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Enhanced identification of defective pre-coated paints on metal through simple temperature-perturbed infrared measurement in conjunction with two-trace two-dimensional correlation analysis

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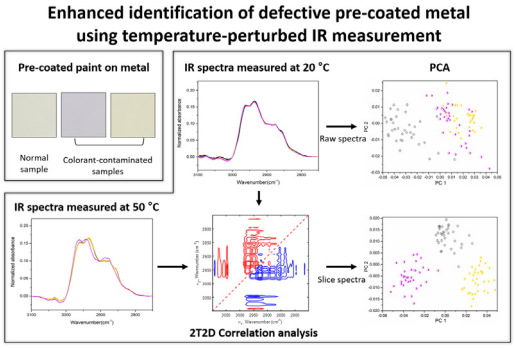
HIGHLIGHTS

- Temperature-perturbed IR measurement was proposed to identify defective pre-coated paints.
- It was to induce dissimilar temperature-driven structural variations of paint and contaminants.
- To effectively recognize the dissimilar behaviors, 2T2D correlation analysis was adopted.
- When IR sample spectra at either 20 or 50 °C were used, the defect identification was inferior.
- 2T2D correlation analysis of both spectra enabled the more accurate defect identification.

GRAPHICAL ABSTRACT



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ABSTRACT

A promising infrared (IR) spectroscopic method able to effectively identify defective pre-coated metal (PCM), a pre-painted metal panel, has been demonstrated. A temperature-perturbed IR measurement in conjunction with a two-trace two-dimensional (2T2D) correlation analysis was proposed as a strategy for enhancing defect identification. Our objectives were to induce dissimilar temperature-driven structural variations of base paints and added components, to recognize dissimilarities by 2T2D correlation analysis, and to use subsequent 2T2D correlation features to identify sample defects. For the exploratory examination, three defect cases were studied: 1) grey-silver PCMs with and without phosphate epoxy (2.0%), 2) normal and violet colorant-contaminated (0.2%) black PCMs, and 3) normal, violet (0.5%), and yellow colorant-contaminated (0.1%) white PCMs. The IR spectral features of the PCMs collected at 20 and 50 °C were different due to the temperature-dependent structural variations. Initial measurements at 50 °C allowed discrimination of normal and violet colorant-contaminated black PCMs. When using 2T2D slice spectra obtained from 2T2D correlation analysis using the spectra measured at the two temperatures, violet- as well as yellow colorant-contaminated white PCMs were identified, while these were unclear in the measurements at either 20 or 50 °C. The effective capture of dissimilar temperature-driven spectral variations of base paint and colorants (contaminants) by 2T2D correlation analysis was responsible for the improved defect identification.

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1. Introduction

Pre-coated metal (PCM) [1–4], a type of pre-painted metal panel, is widely used for the manufacture of refrigerators, washing machines, and microwave ovens. Since PCM is easily processed into the intended shape of a target product, the subsequent manufacturing process is fast and advantageous for mass production. Also, the use of PCM enhances the appearance of products by providing a smooth texture and finished look. Therefore, quality assurance of PCM during production has become important to ultimately ensure the high functionality and high aesthetic value of final PCM-based products. In this context, identification of faulty (defective) PCMs is one of the most demanding tasks. Defective PCMs can result from omission of a necessary functional additive or mistaken addition of unnecessary colorant in PCM paints. There is need for a simple analytical method enabling rapid identification of a defective product.

Infrared (IR) and Raman spectroscopy are potential candidates to meet this demand since they provide rich information on molecular structures of components contained in a sample, and the corresponding sample measurements are fast and non-destructive. Due to these merits, the two spectroscopic methods have been frequently chosen in the field of paint analysis such as for forensic investigations [5–9], assessment of painted art materials [10], and discrimination of commercial paints produced by different manufacturers using chemometrics [11,12]. In this study, IR spectroscopy was adopted to identify defective PCMs, and temperature-varied IR measurements were proposed to enhance identification performance. When the temperature of a coated paint varies, the temperature-driven structural factors of a base paint and an added material (contaminant) differ. Therefore, at a certain temperature, the spectral features of the contaminant can become dissimilar from those of the base paint, leading to more sensitive recognition of defective PCMs. To explore the proposed strategy, three types of defective PCM samples (refer to Fig. 1 as well as the experimental section) were employed and their attenuated total reflection (ATR) IR spectra of the samples acquired at 20 and 50 °C were comparatively analyzed. Also, principal component analysis (PCA) was employed to simply evaluate discriminability of defective samples in a principal component (PC) score domain. Further, two-trace two-dimensional (2T2D) correlation analysis able to sensitively recognize asynchronous spectral behaviors such as dissimilar temperature-driven spectral features in a pair of compared spectra [13–17] was evaluated as a potential tool for better recognition of the dissimilar spectral features and improved identification of defective samples.

2. Experimental section

Three types of defective PCM samples: 1) grey-silver PCMs with and without an adhesion enhancer (phosphate epoxy: 2.0%), 2) normal and violet colorant-contaminated (0.2%) black PCMs, and 3) normal, violet colorant-contaminated (0.5%), and yellow colorant-contaminated (0.1%) white PCMs were provided by KCC Corporation, Korea. An image of the real grey-silver PCM (top right) and the surface colors of the PCMs are shown in Fig. 1. As seen, the PCM has a clean surface without visually observable defects or contaminants. For each case, 7 normal and 7 defective samples were analyzed. ATR IR spectra (4000–800 cm^{-1} range, resolution: 4 cm^{-1}) of the samples were acquired at 20 °C with 64 repeating scans using a Lumos IR spectrometer (Brucker, Germany) equipped with a microscope incorporating a Ge ATR crystal (100 μm) and mercury cadmium telluride (MCT) detector. For each sample, 5 IR spectra were collected by randomly locating and contacting the

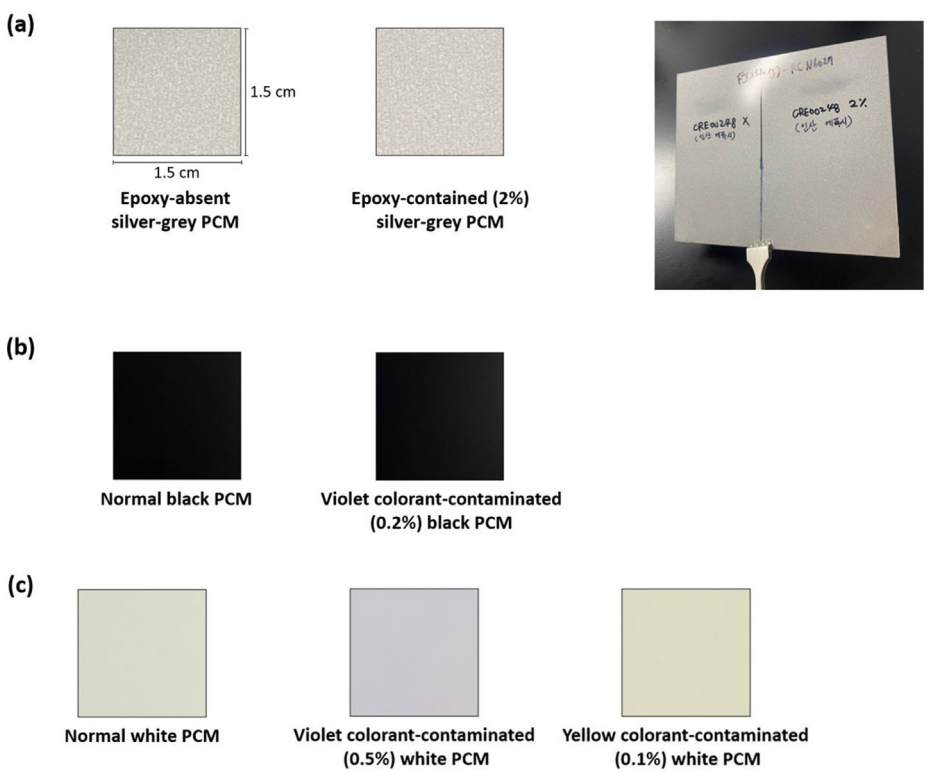

Fig. 1. The surface colors of grey-silver PCMs with and without phosphate epoxy (a), normal and violet colorant-contaminated black PCMs (b), and normal, violet colorant-contaminated, and yellow colorant-contaminated white PCMs (c). A picture of the real grey-silver PCM is shown (top right).

ATR probe on the sample surface. To control the temperature of each sample, each PCM plate was loaded onto a flat voltage-controllable silicon rubber heating pad (FLD heater, China) with firm contact, and the temperature on the PCM surface was verified (±0.2 °C) using a non-contact IR thermometer. Then, PCA was performed using the collected spectra, and the ability to identify defective samples in a principal component (PC) score domain was evaluated.

Next, IR spectra of the #2 and #3 PCM samples were acquired at an elevated temperature of 50 °C. For each case, subsequent two-group discriminability in a PC score domain was compared with that of the 20 °C measurement. Further, 2T2D correlation analysis was performed using the #3 PCM sample spectra measured at 20 °C and 50 °C. PCA was executed using the acquired 2T2D asynchronous correlation slice spectra to identify the defective samples in a PC score domain. Baseline correction of raw spectra based on AsLS (Asymmetric Least Squares smoothing), normalization of baseline-corrected spectra, PCA, and 2T2D correlation analysis were performed using MATLAB version 2020a (The Math Works INC., MA, USA).

3. Results and discussion

3.1. Investigation of grey-silver PCMs with and without phosphate epoxy

One key additive in a PCM is an adhesion enhancer, which increases the adhesion strength of the paint to the metal surface. Therefore, confirmation of additive inclusion in a PCM is required to assure durability of the adhesive on the final product. A phosphate epoxy is a frequently adopted adhesion enhancer. As seen in Fig. 1 (a), the colors of grey-silver PCMs with and without phosphate epoxy (2.0%) are indistinguishable by naked eye. Fig. 2 shows IR spectra of the epoxy-absent (sky blue) and epoxy-added (orange) samples collected at 20 °C in the 3100–2750 (a) and 1800–800 cm^{-1} (b) ranges. As seen, the baselines of spectra varied vertically due to inconsistent contact of the ATR probe with the paint surface due to microscopic surface roughness. The baselines of spectra were corrected using AsLS, and subsequent baseline-corrected spectra were divided by the corresponding peak area for normalization. Fig. 2 (c) and (d) show the average normalized IR spectra of grey-silver PCM

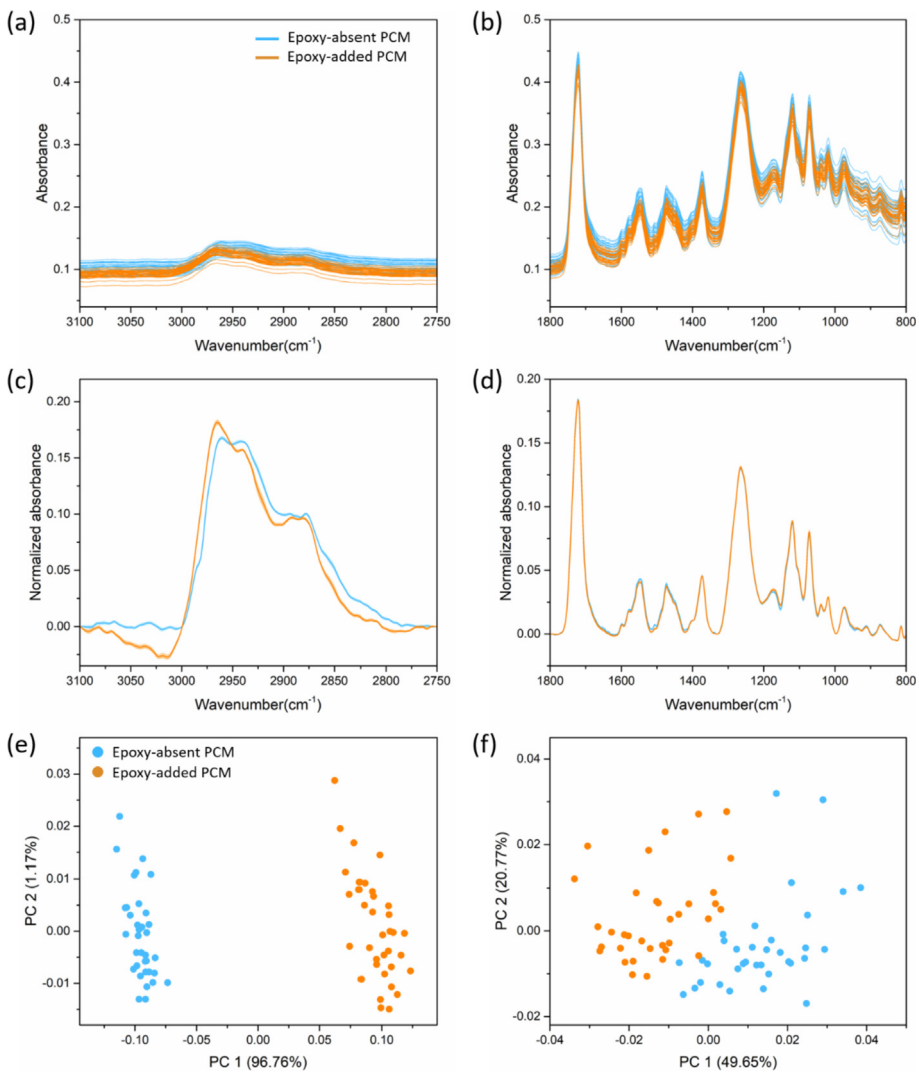

Fig. 2. Raw IR spectra of the epoxy-absent and epoxy-added PCM samples collected at 20 °C in the 3100–2750 (a) and 1800–800 cm^{-1} (b) ranges and the corresponding average normalized spectra ((c), (d)). The shade on each average spectrum indicates the standard deviation of absorbance variation among the involved samples. Distributions of 1st and 2nd PC scores of all the sample spectra in the 3100–2750 (e) and 1800–800 cm^{-1} (f) ranges are shown.

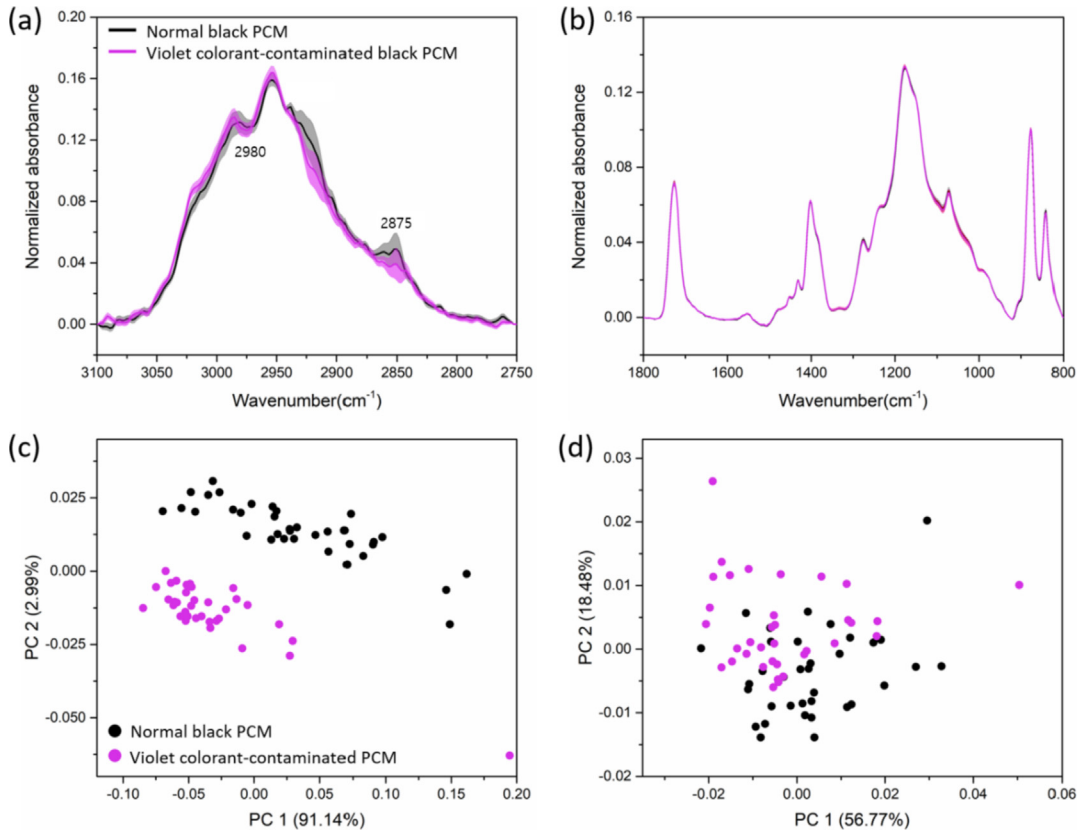

Fig. 3. Average normalized IR spectra of normal and violet colorant-contaminated black PCM samples collected at 20 °C in the 3100–2750 (a) and 1800–800 cm^{-1} (b) ranges. Distribution of 1st and 2nd PC scores of all the sample spectra in the 3100–2750 (c) and 1800–800 cm^{-1} (d) ranges are shown.

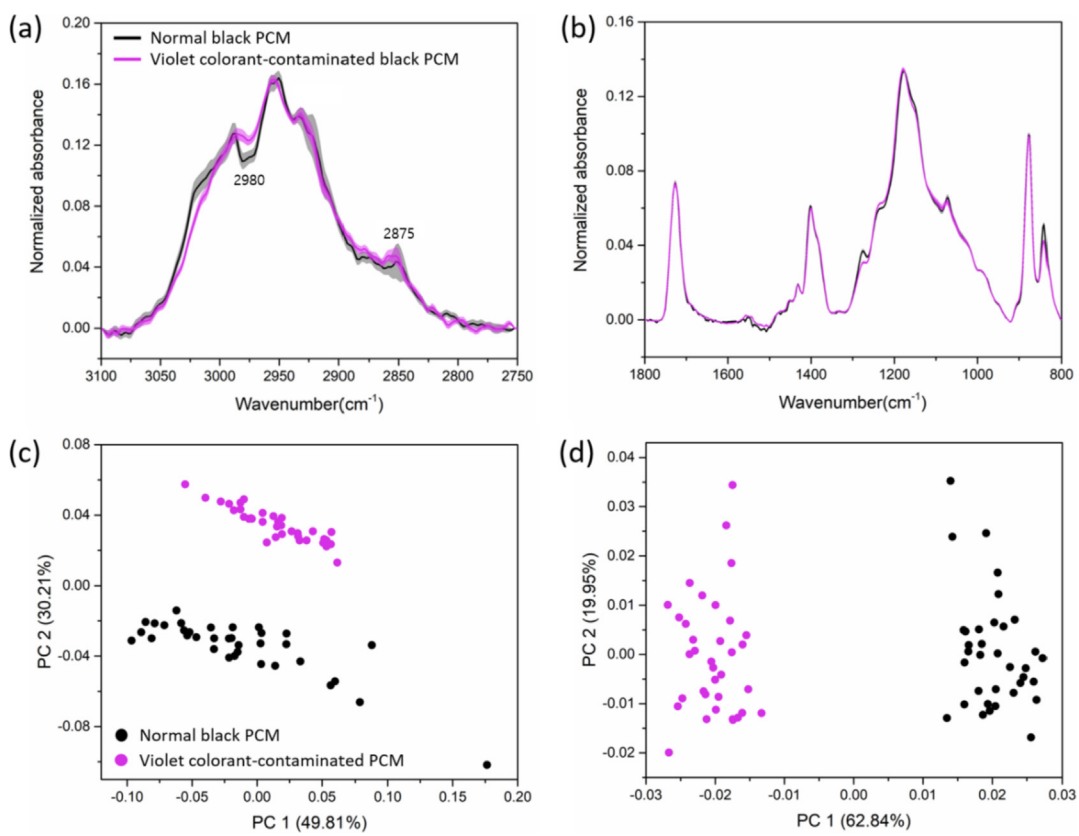

Fig. 4. Average normalized IR spectra of normal and violet colorant-contaminated black PCM samples collected at 50 °C in the 3100–2750 (a) and 1800–800 cm^{-1} (b) ranges. Distribution of 1st and 2nd PC scores of all the sample spectra in the 3100–2750 (c) and 1800–800 cm^{-1} (d) ranges are shown.

samples in the 3100–2750 and 1800–800 cm^{-1} ranges, respectively. The shade on each average spectrum indicates the standard deviation of absorbance variation among the samples. As seen, the peak shapes of samples with epoxy and of samples without epoxy were different in the 3100–2750 cm^{-1} range, since the concentration of added phosphate epoxy was 2.0%. This was large enough to be recognized by IR spectroscopy. Although the spectral features of the two groups in the 1800–800 cm^{-1} range were similar, moderate differences were observed in the 1558–1410, 1205–1146, and 973–885 cm^{-1} ranges.

To easily compare the features of all the spectra, PCA was executed using the normalized spectra, and spreads of subsequent principal component (PC) scores were investigated. Fig. 2 (e) and (f) show the distributions of 1st and 2nd PC scores obtained using the IR information in the 3100–2750 and 1800–800 cm^{-1} ranges, respectively. The percentages of spectral variation explained by the respective PCs are denoted in the *x* and *y* axes. As expected, the scores of the two groups were clearly distinguishable using the 3100–2750 cm^{-1} range. The two groups also were recognizable in the score domain based on the 1800–800 cm^{-1} range, while their separation was not distinct compared to that using the 3100–2750 cm^{-1} range.

3.2. Identification of violet-colorant contamination in black PCM

During production of PCMs, addition of unnecessary colorant in paint occasionally occurs. The normal and violet colorant (0.2%)-contaminated black PCMs were adopted as a typical case. Again, the colors of the two samples were indistinguishable by naked eye observation (Fig. 1 (b)). Fig. 3 displays the average normalized IR spectra of normal (black) and violet colorant-contaminated (violet) black PCM samples collected at 20 °C in the 3100–2750 (a) and 1800–800 cm^{-1} (b) ranges. The shade on each average spectrum indicates the standard deviation of absorbance variation. The spectral features of the two samples in the 3100–2750 cm^{-1} range were similar since the contamination level was only 0.2%. Nonetheless, considerable differences were observed around the peaks at 2980 and 2875 cm^{-1}. In contrast, the peak shapes of the samples in the 1800–800 cm^{-1} range were similar without distinct differences. Therefore, the two groups were separated in the score domain based on the 3100–2750 cm^{-1} range (Fig. 3 (c)), while they were inseparable using the 1800–800 cm^{-1} range (Fig. 3 (d)) due to the considerable mutual overlap of the scores.

Although identification of colorant-contaminated black PCM using the 3100–2750 cm^{-1} range is feasible, enhancing spectral difference in the 1800–800 cm^{-1} range is beneficial to ensure a more accurate analysis. As discussed in the Introduction, molecular structures of a base paint and a contaminating colorant would vary differently with temperature change of the PCM. Therefore, at a certain temperature, spectral features of a contaminant can be quite dissimilar from those of base paint, and recognition of contaminated PCMs would become easier. Based on this possibility, IR spectra of the same samples were collected at an elevated temperature of 50 °C. The resulting average spectra in the 3100–2750 and 1800–800 cm^{-1} ranges are shown in Fig. 4 (a) and (b), respectively. The spectral features of the samples varied considerably compared to those measured at 20 °C, especially in the 3042–2964 and 1295–1048 cm^{-1} ranges. In these ranges, a few peaks in the two groups do not overlap, which was not observed in the 20 °C measurement (Fig. 3). With the 50 °C measurement, the two groups were separated better in the score domain based on the 3100–2750 cm^{-1} range (Fig. 4 (c)), and two-group discrimina-

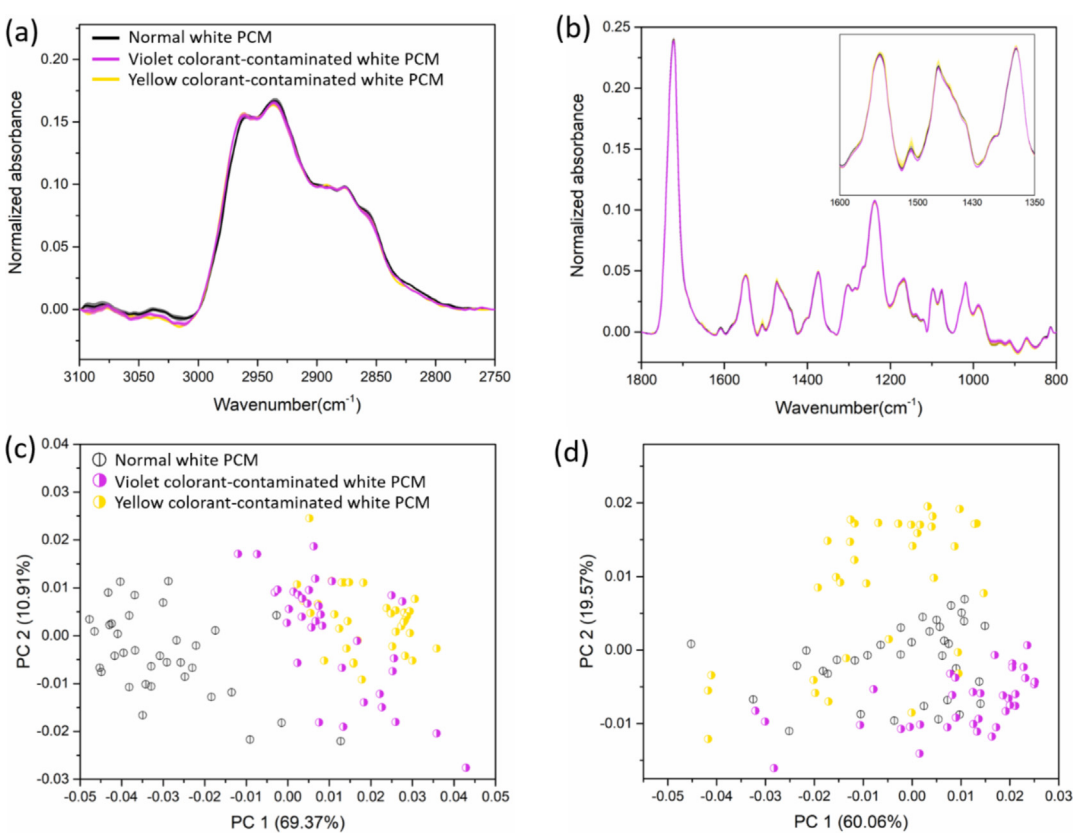

Fig. 5. Average normalized IR spectra of normal, violet colorant-contaminated, and yellow colorant-contaminated white PCM samples collected at 20 °C in the 3100–2750 (a) and 1800–800 cm^{-1} (b) ranges. The inset is magnification of the 1600–1350 cm^{-1} range. Distributions of 1st and 2nd PC scores of all the sample spectra in the 3100–2750 (c) and 1800–800 cm^{-1} (d) ranges are shown.

tion was apparent in the 1800–800 cm^{-1} range (Fig. 4 (d)). The dissimilar temperature-induced spectral variations of the base paint and contaminant at 50 °C enabled clear recognition of the contaminated samples even using the corresponding IR information in the 1800–800 cm^{-1} range.

3.3. Discrimination of normal, violet colorant-, and yellow colorant-contaminated white PCMs

To further explore the potential of temperature-perturbed IR analysis, white PCM samples separately contaminated with a violet colorant (0.1%) or a yellow-colorant (0.5%) were prepared, and identification of these contaminants were attempted. The surface colors of the samples are presented in Fig. 1 (c). Fig. 5 shows the average IR spectra of normal (black), violet colorant-contaminated (violet), and yellow colorant-contaminated (yellow) white PCM samples collected at 20 °C in the 3100–2750 (a) and 1800–800 cm^{-1} (b) ranges. In the 3100–2750 cm^{-1} range, the overall spectral features of the three sample groups were similar to each other. The peak shapes of the normal sample were slightly different from those of the contaminated samples, as in the 3066–3003 and 2838–2800 cm^{-1} ranges, though significant differences between the two contaminated sample spectra were not observed. The spectra of three groups did not have noticeable differences in the 1800–800 cm^{-1} range. Peaks in the 1600–1350 cm^{-1} range are magnified in the inset and show no noticeable spectral differences among the three groups. In the score domain based on the 3100–2750 cm^{-1} range (Fig. 5 (c)), the normal samples were moderately separated from the two contaminated sample groups, while contaminated sample groups could not be discriminated. In the 1800–800 cm^{-1} range, the three groups were recognizable (Fig. 5 (d)) but did extensively and mutually overlap without clear boundaries.

Fig. 6 (a) and (b) present the average IR spectra of the same samples collected at 50 °C in the 3100–2750 and 1800–800 cm^{-1} ranges, respectively. Notably, the spectral features of violet colorant-contaminated sample in the 3100–2750 cm^{-1} range varied significantly compared to those acquired at 20 °C (Fig. 5 (a)), while the spectral shapes of the remaining two groups were similar. Therefore, the violet colorant-contaminated samples were clearly separated from the rest of the samples in the score domain constructed using the 3100–2750 cm^{-1} (Fig. 6 (c)), while the normal and yellow colorant-contaminated samples were inseparable. Also, the spectral features of the samples in the 1800–800 cm^{-1} range changed slightly with increased temperature, specifically over the 1600–1350 cm^{-1} range (refer to the inset). In comparison with the results of the 20 °C measurement (Fig. 5 (d)), the three groups were differentiated better in the corresponding score domain (Fig. 6 (d)). However, their clear separation was not realized.

Instead of separately using the IR information obtained at either 20 or 50 °C, simultaneous utilization of information at the two temperatures would potentially improve discrimination. For this purpose, 2T2D correlation analysis, which can highlight dissimilar temperature-induced spectral features in a pair of compared spectra [13], was adopted, and the spectra of each sample collected at 20 and 50 °C were used together for the analysis. The IR spectra acquired at these temperatures are referred as the sample $s(\upsilon)$ spectrum (50 °C) and reference $r(\upsilon)$ spectrum (20 °C), respectively. An asynchronous spectrum, $\Psi(\upsilon_1, \upsilon_2)$, which is a quantitative comparison of the spectral intensity variations observed in two spectral variables, was obtained using equation (1).

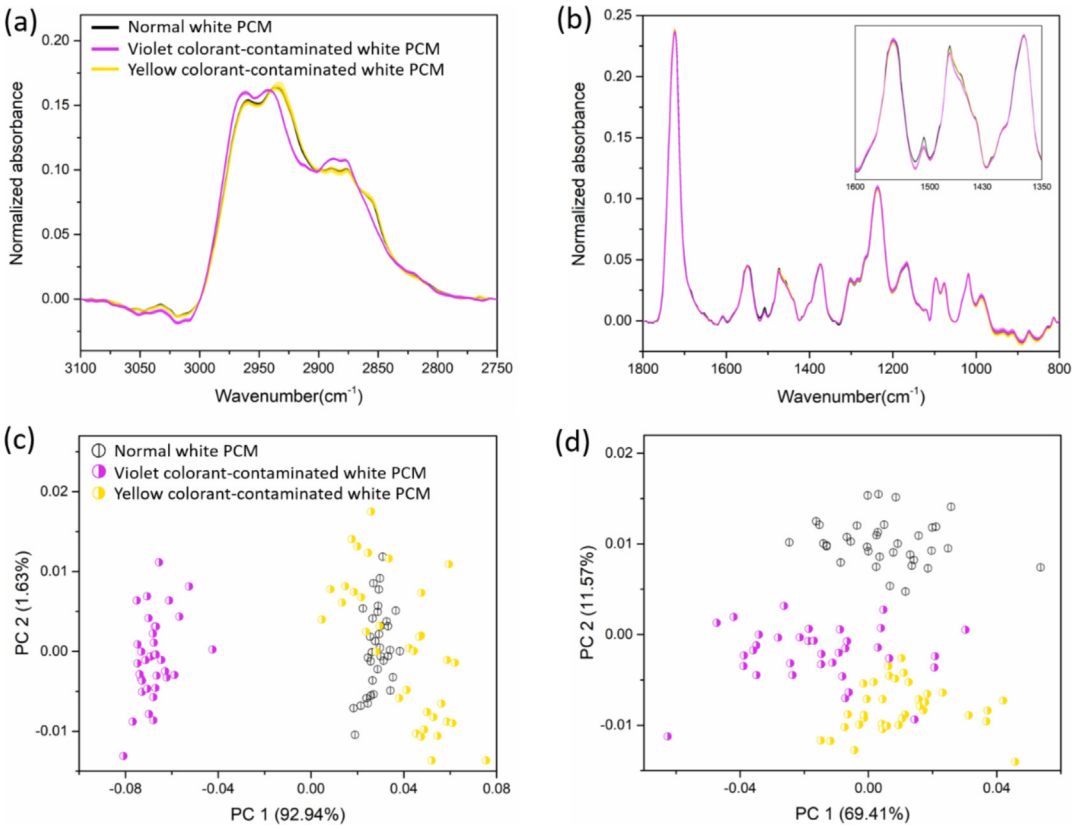

Fig. 6. Average normalized IR spectra of normal, violet colorant-contaminated, and yellow colorant-contaminated white PCM samples collected at 50 °C in the 3100–2750 (a) and 1800–800 cm^{-1} (b) ranges. The inset shows a magnified version of the 1600–1350 cm^{-1} range. Distributions of 1st and 2nd PC scores of all the sample spectra in the 3100–2750 (c) and 1800–800 cm^{-1} (d) ranges are shown.

$$\Psi(v_1, v_2) = \frac{1}{2}\left[s(v_1)\cdot r(v_2) - r(v_1)\cdot s(v_2)\right] \quad (1)$$

A detailed description of 2T2D correlation analysis can be found in previous publications. [13,14]

Fig. 7 presents the asynchronous 2T2D correlation plots obtained by 2T2D correlation analysis of the average spectra of normal white PCM collected at 20 and 50 °C in the 3100–2750 (a) and 1800–800 cm^{-1} (b) ranges. The asynchronous 2T2D correlation plots obtained from the violet colorant-contaminated (Fig. 7 (c) and (d)) and yellow colorant-contaminated ((Fig. 7 (e) and (f)) white PCM samples also are shown. The blue and red colors indicate positive and negative correlations, respectively. It is notable that the 2T2D asynchronous correlation patterns in the 3100–3000 cm^{-1} range were substantially different among the three samples. A broad asynchronous correlation appeared over the 3100–3000 cm^{-1} range in the normal sample. This correlation decreased into the 3100–3040 and 3070–3000 cm^{-1} ranges for the violet colorant-contaminated and yellow colorant-contaminated samples, respectively. Also, the asynchronous correlation patterns in the 3000–2750 cm^{-1} range were considerably different among groups. Asynchronous correlation was observed over the 3100–2750 cm^{-1} range for the yellow colorant-contaminated sample, while it narrowed to the 3000–2850 cm^{-1} range for the normal sample. For the violet colorant-contaminated sample, the positive and negative correlations showed more repetition over the 3000–2800 cm^{-1} range, indicating temperature-induced peak shifts at narrow locations. Overall, the dissimilar asynchronous correlation patterns originated from the included colorants, which have temperature-dependent behaviors different from those of the base white paint. Meanwhile, the asynchronous correlation patterns in the 1800–800 cm^{-1} range were slightly different from each other, but not as distinctive as those in the 3100–2750 cm^{-1} range.

The 2T2D asynchronous correlation maps in the 3100–2750 and 1800–800 cm^{-1} ranges consisted of 172 and 491 slice spectra, respectively. All the slice spectra in each spectral range were serially connected, and the merged 2T2D slice spectra were subject to PCA. Fig. 8 (a) and (b) show the score plots obtained using the merged 2T2D slice spectra in the 3100–2750 and 1800–800 cm^{-1} ranges, respectively. Interestingly, isolation of three groups became apparent without inter-group overlapping in the 3100–2750 cm^{-1} range. In contrast, clear three-group separation was

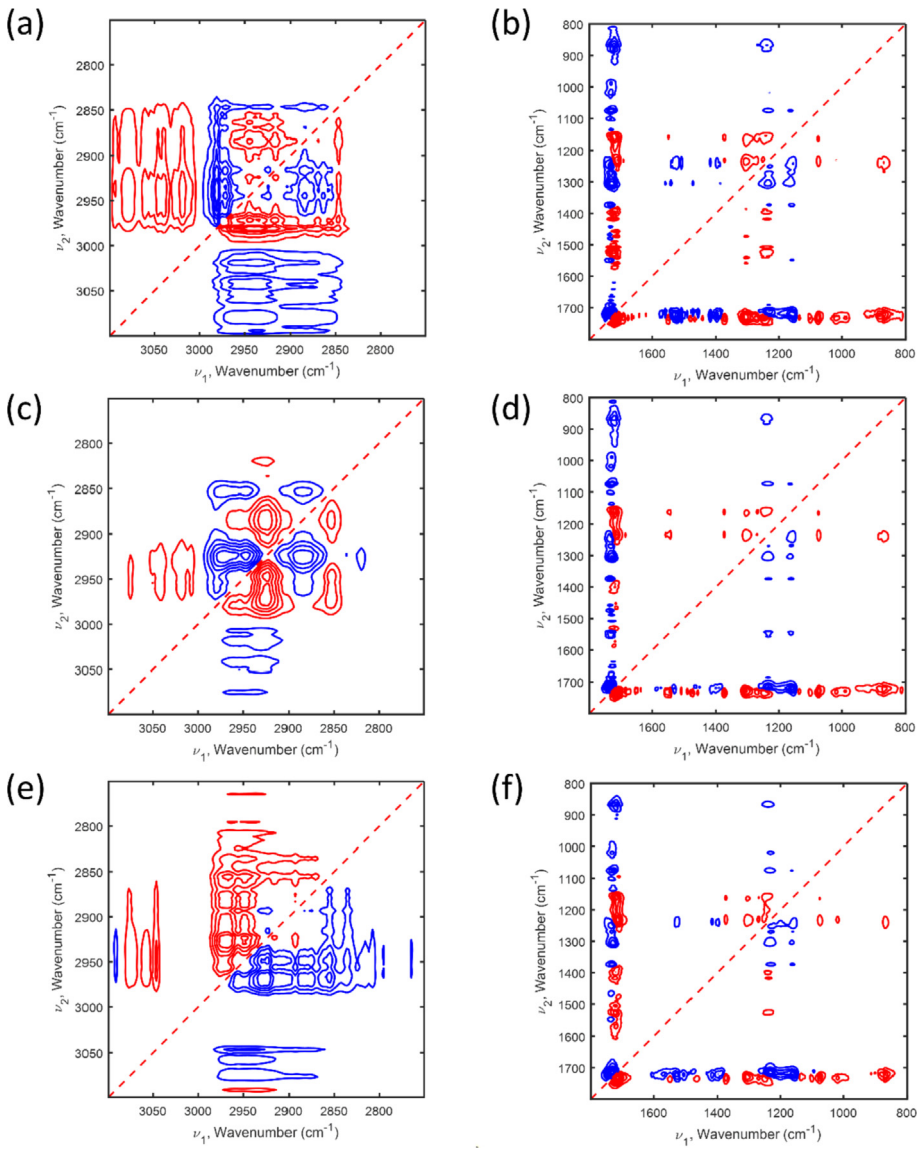

Fig. 7. 2T2D asynchronous correlation maps of normal (first row), violet colorant-contaminated (second row), and yellow colorant-contaminated (third row) white PCM samples acquired in the 3100–2750 (left column) and 1800–800 cm^{-1} (right column) ranges.

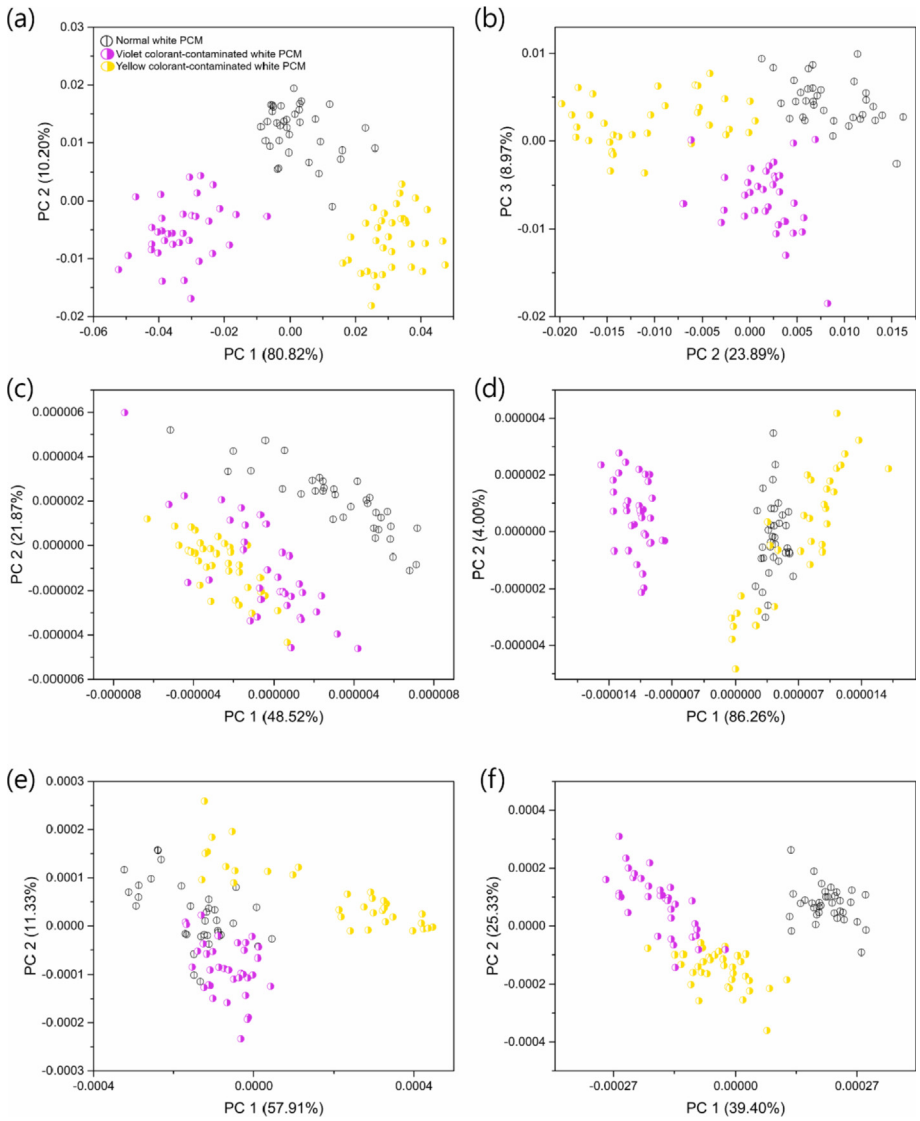

Fig. 8. The score plots obtained using the merged 2T2D slice spectra in the 3100–2750 (a) and 1800–800 cm^{-1} (b) ranges resulted from 2T2D correlation analysis using the sample spectra collected at 20 and 50 °C. The score plots acquired using the merged 2T2D slice spectra in the 3100–2750 (c) and 1800–800 cm^{-1} (d) ranges resulted from 2T2D correlation analysis using the sample spectra collected at 20 °C with the average spectrum of normal samples at the same temperature as the reference are also shown. The results of same constant-temperature measurement at 50 °C are also displayed ((e) and (f) using the 3100–2750 and 1800–800 cm^{-1} range, respectively).

not realized in the measurements at 20 and 50 °C (Fig. 5 (c) and Fig. 6 (c), respectively). This result demonstrates the utility of 2T2D correlation analysis for recognition of dissimilar temperature-induced spectral features of the base paint and colorants, making it an effective tool for identification of colorant-contaminated samples. Although the three-group discrimination using the 1800–800 cm^{-1} range (Fig. 8 (b)) was less apparent, it is superior to those from measurements at 20 and 50 °C (Fig. 5 (d) and Fig. 6 (d), respectively).

Meanwhile, a strategy of employing the average spectrum of one sample group as the reference spectrum in the 2T2D correlation analysis would be simple and alternatively versatile. To evaluate utility of this method, the average spectrum of normal white PCM sample spectra collected at 20 °C was used as the reference spectrum and 2T2D correlation analysis was performed for all the normal, violet colorant-contaminated, and yellow colorant-contaminated white PCM sample spectra collected at 20 °C. Fig. 8 (c) and (d) presents the score plots obtained using the merged 2T2D slice spectra in the 3100–2750 and 1800–800 cm^{-1} ranges, respectively. In the score domain based on the 3100–2750 cm^{-1} range (Fig. 8 (c)), the normal samples are moderately separated from the two contaminated sample groups, while the contaminated sample groups could not be discriminated. Meanwhile, in the score domain based on the 1800–800 cm^{-1} range (Fig. 8 (d)), the violet colorant-contaminated samples are isolated from the remained sample groups, while the normal and yellow colorant-contaminated sample groups are inseparable.

Also, 2T2D correlation analysis was performed for all the white PCM sample spectra collected at 50 °C using the average spectrum of normal sample spectra collected at 50 °C as the reference. Fig. 8 (e) and (f) displays the score plots obtained using the merged 2T2D slice spectra in the 3100–2750 and 1800–800 cm^{-1} ranges, respectively. In the score domain based on the 3100–2750 cm^{-1} range (Fig. 8 (e)), the yellow colorant-contaminated samples are somewhat away from the remained samples, while the normal and violet colorant-contaminated samples are considerably overlapped. With using the 1800–800 cm^{-1} range (Fig. 8 (f)), the normal samples are separable from the contaminated sample groups, while the two contaminated sample groups could not be differentiated.

In overall, the discrimination performances with adopting the average spectrum of normal sample group as the reference spectrum in the 2T2D correlation analysis were inferior to those with using two spectra of the sample measured at 20 and 50 °C. This comparison again supports that the temperature-driven structural variations of the base paint and added contaminants are different and more informative to identify the defective (contaminated) samples with using 2T2D correlation analysis.

4. Conclusion

Temperature-varied IR measurements were used to enhance the identification of defective black PCMs. Moreover, when combined with 2T2D correlation analysis, identification of violet colorant-contaminated and yellow colorant-contaminated white PCM samples was substantially improved. Enhanced identification was attributed to generation of dissimilar spectral variations of the base paints and colorants by temperature and sensitive recognition of dissimilarities by 2T2D correlation analysis. Conventional two-dimensional (2D) correlation analysis based on a series of temperature-varied spectra (more than 3 spectra) could be also considered for the same analysis; while, 2T2D correlation analysis based on the direct comparison of two temperature-varied spectra, instead of trend-tracking in a series of temperature-varied spectra in 2D correlation analysis, expects to more sensitively recognize the temperature-induced spectral variation, as demonstrated in a former publication that compared both methods for the near-infrared spectroscopic authentication of olive oil [16]. Moreover, since 2T2D correlation analysis requires only two spectra, subsequent acquisition of temperature-varied spectra is faster and easier.

By the way, an IR measurement at 50 °C was attempted in this study as an exploratory investigation. However, an optimal temperature yielding superior discrimination performance (with different temperature-driven spectral features) probably exists for a given sample based on the results in previous publications [18–20]. Each optimal temperature needs to be identified when the demonstrated analytical scheme is implemented at a real QC lab. Also, advanced and more powerful classification methods such as support vector machine (SVM) could be beneficial for discriminant analysis for sample groups with adjacent boundaries.

CRediT authorship contribution statement

Youngtaek Ma: Conceptualization, Investigation, Resources. **Daun Seol:** Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing - review & editing, Visualization. **Kyungmin Nam:** Software, Validation, Formal analysis, Investigation, Data curation, Writing - review & editing, Visualization. **Woosuk Sohng:** Software, Data curation. **Hoeil Chung:** Conceptualization, Methodology, Resources, Writing - original draft, Writing - review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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